

Lecture 6

Vector calculus

$f: \mathbb{R}^n \rightarrow \mathbb{R}$, then

$$\int_{\mathbb{R}^d} f dx = \int_{-\infty}^{\infty} \left(\dots \left(\int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(x_1, \dots, x_n) dx_1 \right) dx_2 \right) \dots \right) dx_n$$

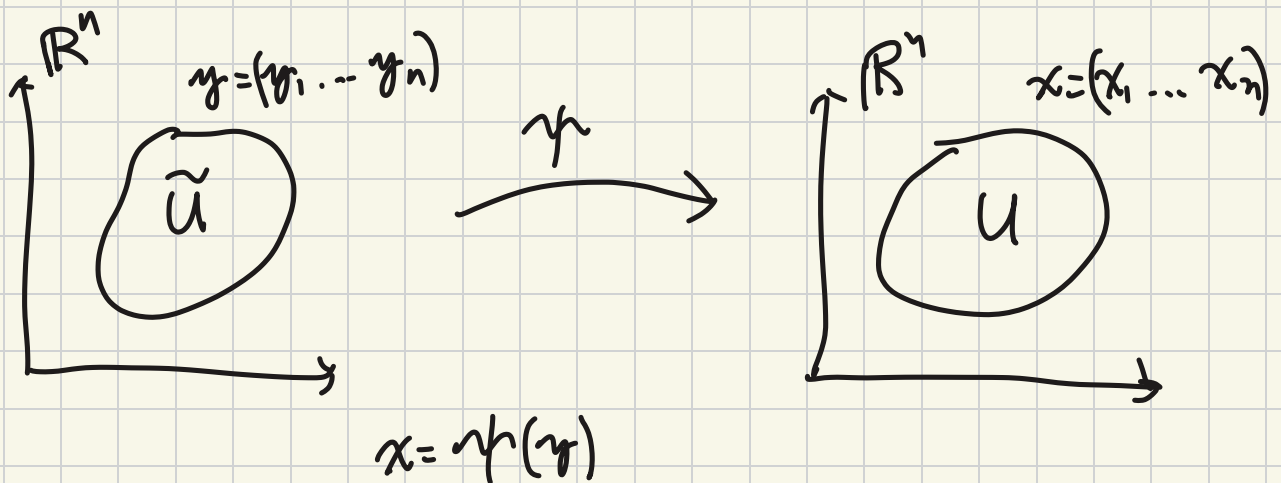
If $U \subset \mathbb{R}^n$ and $f: U \rightarrow \mathbb{R}$, then

$$\int_U f dx = \int_{\mathbb{R}^d} \mathbb{1}_U(x) f(x) dx$$

where $\mathbb{1}_U(x) = \begin{cases} 1 & x \in U \\ 0 & x \notin U \end{cases}$ (indicator function)

Change of variables

$$\gamma: \tilde{U} \rightarrow U, \quad \tilde{U}, U \subset \mathbb{R}^n$$



$f: U \rightarrow \mathbb{R}$ pulls back to $f \circ \gamma: \tilde{U} \rightarrow \mathbb{R}$

In other words, we changed variables from $f(x)$ to $f(x(y))$

Recall $D\gamma = \frac{\partial x}{\partial y} = \begin{pmatrix} \frac{\partial x_1}{\partial y_1} & \dots & \frac{\partial x_1}{\partial y_n} \\ \vdots & & \vdots \\ \frac{\partial x_n}{\partial y_1} & \dots & \frac{\partial x_n}{\partial y_n} \end{pmatrix}$

Write $\left| \frac{\partial x}{\partial y}(\gamma) \right| = \left| \det \frac{\partial x}{\partial y}(\gamma) \right|$

Change of variable formula:

$$\int_U f(x) dx = \int_{\tilde{U}} f(x(\gamma)) \left| \frac{\partial x}{\partial y}(\gamma) \right| d\gamma$$

Ex 1. $\int_0^1 f(2y) dy$ $x=2y$

$$= \int_0^1 \frac{1}{2} f(2y) \cdot 2 dy$$

$$= \frac{1}{2} \int_0^2 f(x) dx$$

Ex 2. Polar coordinates

$$x(r, \theta) = r \cos \theta$$

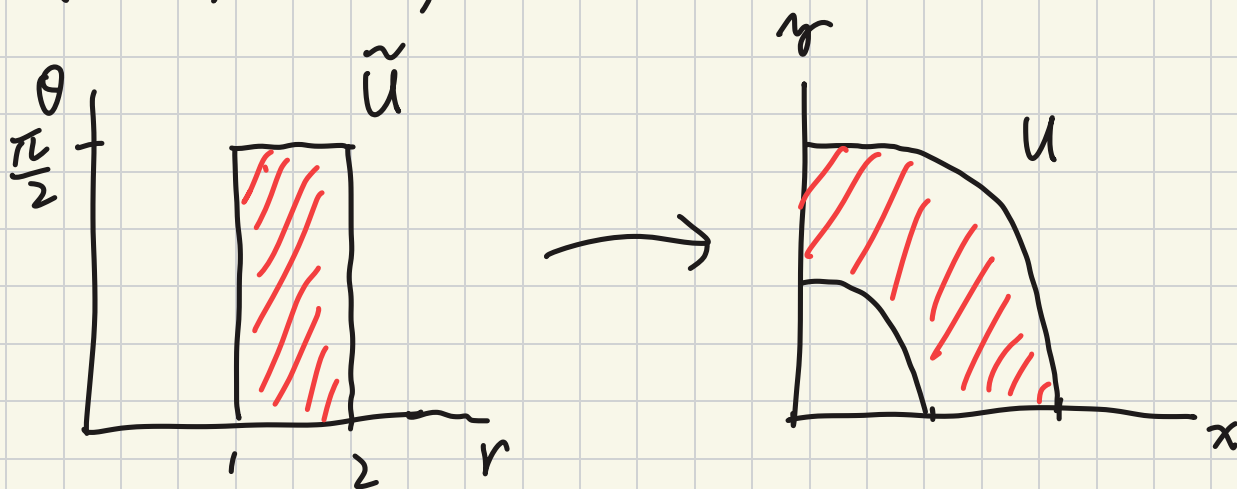
$$y(r, \theta) = r \sin \theta$$

$$\begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}$$

$$\Rightarrow \frac{\partial(x,y)}{\partial(r,\theta)} = r$$

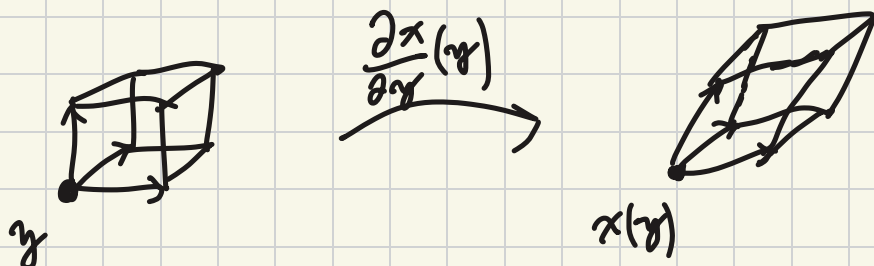
$$\int_U f(x,y) dx dy = \int_{\tilde{U}} \tilde{f}(r,\theta) r dr d\theta$$

where $\tilde{f}(r,\theta) = f(x(r,\theta), y(r,\theta))$
 by an abuse of notation, we often write
 $\tilde{f}(r,\theta) = f(r,\theta)$



$$\int_U dx dy = \int_0^{\frac{\pi}{2}} \int_1^2 r dr d\theta = \frac{\pi}{2} \left(\frac{1}{2}(2)^2 - \frac{1}{2}(1)^2 \right) = \frac{3\pi}{4}$$

Justification: $\frac{\partial x}{\partial y}(y)$ is the linear approx near y



infinitesimal volume $dy = \left| \frac{\partial x}{\partial y} \right| dx$

Integration on k -dim surface

$\Gamma \subset \mathbb{R}^n$ a k -dim surface, i.e. $\exists$ open

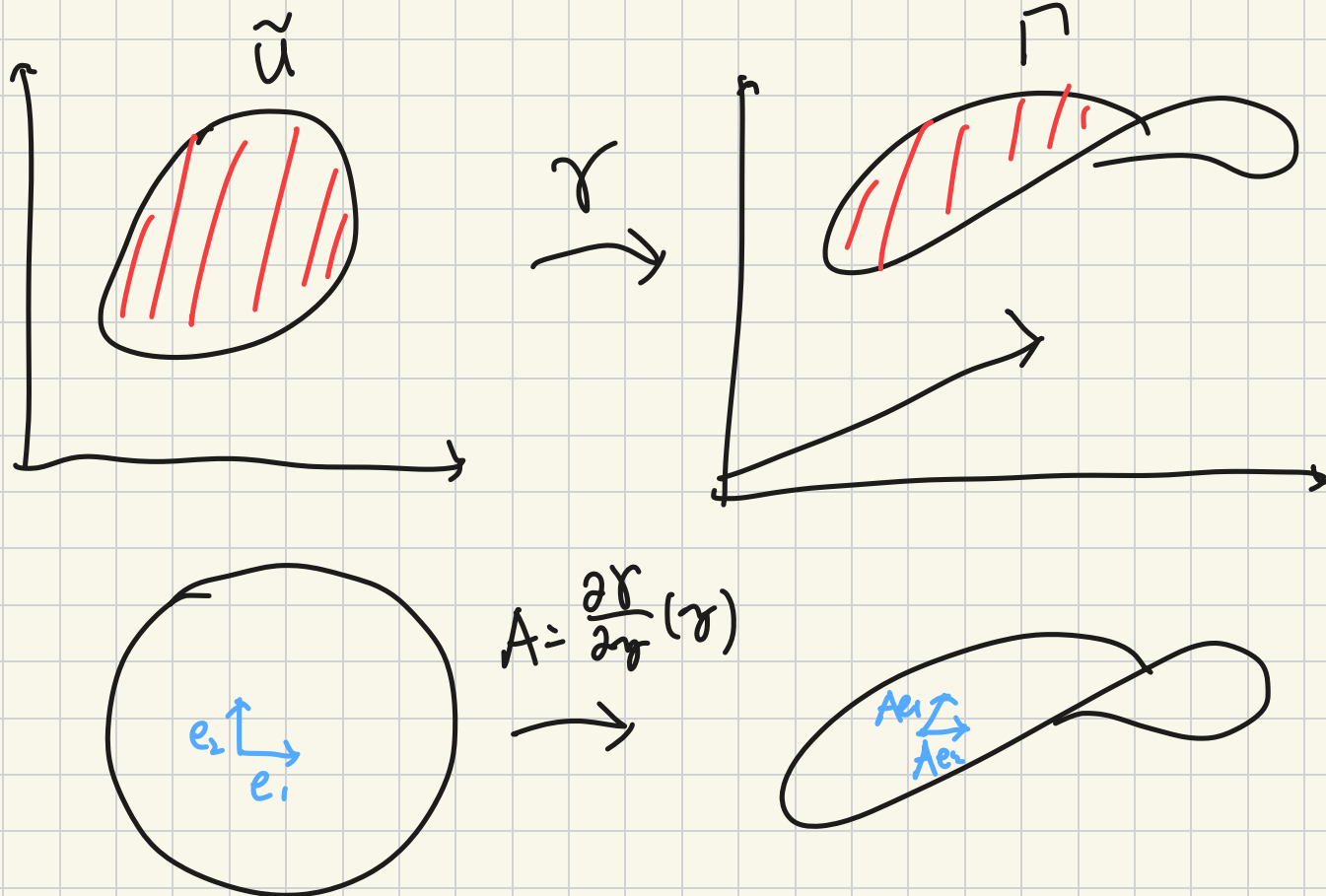
$\tilde{U} \subseteq \mathbb{R}^k$ and a map $\gamma: \tilde{U} \rightarrow \Gamma$ in C^1
s.t. $\frac{\partial \gamma}{\partial y}(\gamma)$ is full rank at every $y \in \tilde{U}$.

γ is called a proper parameterization

Then

$$\int_{\Gamma} f(x) dS(x) := \int_{\tilde{U}} f(\gamma(y)) \left| \frac{\partial \gamma}{\partial y}(\gamma) \right| dy$$

where $\left| \frac{\partial \gamma}{\partial y}(\gamma) \right| = \text{volume of parallelepiped spanned by columns.}$



Example 1 curves $I \subset \mathbb{R}$, $\gamma: I \rightarrow \mathbb{R}^n$

$$\Gamma = \gamma(I)$$

$$\int_{\Gamma} f(x) dS(x) = \int_I f(\gamma(y)) |\gamma'(y)| dy$$

Example 2 Surface in $\mathbb{R}^3$, $U \subset \mathbb{R}^2$

$$\gamma: U \rightarrow \Gamma \subset \mathbb{R}^3$$

$$\frac{\partial \gamma}{\partial y}(y) = \left(\frac{\partial \gamma}{\partial y_1}, \frac{\partial \gamma}{\partial y_2} \right),$$

$$\frac{\partial \gamma}{\partial y_i} = \begin{pmatrix} \frac{\partial \gamma_1}{\partial y_i} \\ \frac{\partial \gamma_2}{\partial y_i} \\ \frac{\partial \gamma_3}{\partial y_i} \end{pmatrix}$$

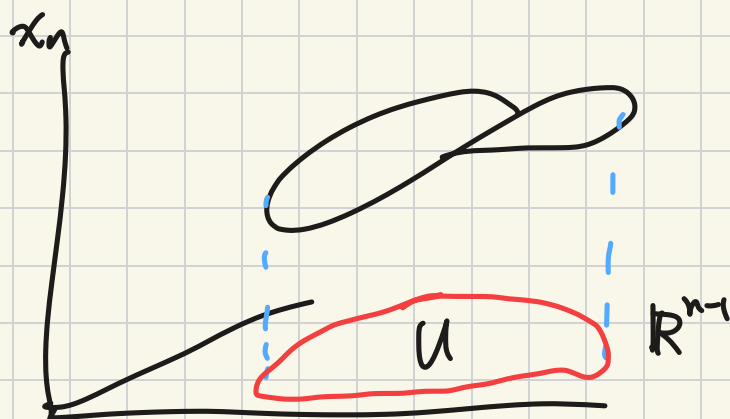
$$\left| \frac{\partial \gamma}{\partial y} \right| = \left| \frac{\partial \gamma}{\partial y_1} \times \frac{\partial \gamma}{\partial y_2} \right|$$

$$\Rightarrow \int_{\Gamma} f(x) dS(x) = \int_U f(\gamma(y)) \left| \frac{\partial \gamma}{\partial y_1}(y) \times \frac{\partial \gamma}{\partial y_2}(y) \right| dy$$

Ex 3 Graphs/hypersurface.

Γ given by graph of $x_n = h(x')$, $x' = \begin{pmatrix} x_1 \\ \vdots \\ x_{n-1} \end{pmatrix}$
parameterized by

$$\gamma(y) = (y, h(y)) \quad y \in U \subset \mathbb{R}^{n-1}$$



Need volume of parallelepiped spanned by

$$\left(\frac{\partial \gamma}{\partial y_1}, \dots, \frac{\partial \gamma}{\partial y_{n-1}} \right) = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \\ \frac{\partial y_1 h}{\partial y_1} & \frac{\partial y_1 h}{\partial y_2} & \dots & \frac{\partial y_1 h}{\partial y_{n-1}} \end{pmatrix}$$

equal to volume spanned by

$$\left(\frac{\partial \gamma}{\partial y_1}, \dots, \frac{\partial \gamma}{\partial y_{n-1}}, \nu \right)$$

where ν is the normal vector.

Note that
$$J(y) = \frac{1}{\sqrt{1 + |Dh(y)|^2}} \begin{pmatrix} (Dh(y))^T \\ -1 \end{pmatrix}$$

where
$$Dh = \left(\frac{\partial r}{\partial y_1}, \dots, \frac{\partial r}{\partial y_{n-1}} \right)$$

Need

$$\frac{1}{\sqrt{1 + |Dh(y)|^2}} \left| \det \begin{pmatrix} Id & |Dh|^T \\ Dh & -1 \end{pmatrix} \right|$$

$$= \sqrt{1 + |Dh(y)|^2} \quad (\text{exercise by induction}).$$

Spherical decomposition. $x \in \mathbb{R}^n$, $r = |x|$

$$\mathcal{S}^{n-1} := \{r=1\} \subset \mathbb{R}^n$$

$x \neq 0$ uniquely written as

$$x = rw, \quad w \in \mathcal{S}^{n-1}, \quad r > 0$$

Rmk. $\Omega \subset \mathbb{R}^n$, $\partial\Omega$ is a hyper surface, we say it is C^1 if it can be written as the graph of a C^1 function

Let $w(y)$ be a parameterization of S^{n-1}
by coordinates $y \in U \subset \mathbb{R}^{n-1}$

C.O.V. $(r, y) \mapsto x = rw(y)$

$$\begin{aligned} dx &= \left| \det \left(\frac{\partial x}{\partial y_1}, \dots, \frac{\partial x}{\partial y_{n-1}}, \frac{\partial x}{\partial r} \right) \right| dr dy_1 \dots dy_{n-1} \\ &= \left| \det \left(\frac{\partial w}{\partial y_1}, \dots, \frac{\partial w}{\partial y_{n-1}}, w \right) \right| r^{n-1} dr dy_1 \dots dy_{n-1} \end{aligned}$$

Observe that w = unit outward normal vector on S

$$\Rightarrow dx = r^{n-1} dr dS(w)$$

Upshot. integrating over ball $B(0; R)$

$$\begin{aligned} \int_{B(0, R)} f(x) dx &= \int_{S^{n-1}} \int_0^R f(rw) r^{n-1} dr dS(w) \\ &= \int_0^R \int_{\partial B(0, r)} f(x) dS(x) dr \end{aligned}$$

Thm (Divergence Thm) Suppose $\Omega \subset \mathbb{R}^n$ is a bdd domain w/ C^1 boundary. For a vector field $F \in C^1(\bar{\Omega}; \mathbb{R}^n)$,

$$\int_{\Omega} \nabla \cdot F \, dx = \int_{\partial\Omega} F \cdot \nu \, dS$$

where ν is the outward pointing normal.

Ex 1. $F = \nabla u$ for some $u \in C^2(\mathbb{R}^n)$

$$\int_{\Omega} \Delta u \, dx = \int_{\Omega} \nabla \cdot F \, dx$$

$$= \int_{\partial\Omega} F \cdot \nu \, dS$$

$$= \int_{\partial\Omega} \nabla u \cdot \nu \, dS$$

$$= \int_{\partial\Omega} D_{\nu} u \, dS = \int_{\partial\Omega} \frac{\partial u}{\partial \nu} \, dS$$

Ex 2. Assume $u \in C^2(\mathbb{R}^3)$ is radial, i.e. $u(x) = u(r)$ where $r = |x|$. Then

$$\int_{B(0,a)} \Delta u = \int_{\partial B(0,a)} \frac{\partial u}{\partial \nu} \, dS$$

$$= \int_{\partial B(0,a)} \frac{\partial u}{\partial r} \, dS = 4\pi a^2 \frac{\partial u}{\partial r}(a).$$

On the other hand, by spherical integration

$$\int_{B(0,a)} \Delta u \, dx = 4\pi \int_0^a \Delta u(r) r^2 \, dr$$

$$\Rightarrow \int_0^a \Delta u(r) r^2 \, dr = a^2 \frac{\partial u}{\partial r}(a)$$

$$FTC \Rightarrow a^2 \Delta u(a) = \frac{\partial}{\partial a} a^2 \frac{\partial u}{\partial r}(a)$$

$$\begin{aligned} \Rightarrow \Delta u(r) &= r^{-2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} u(r) \\ &= \left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} \right) u(r) \end{aligned}$$